

ANIRUDH P

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PROFESSIONAL SUMMARY

AI Engineer-focused BCA (AI & ML) graduate with hands-on experience building end-to-end ML systems — from computer vision pipelines (YOLOv8, EasyOCR) to RAG applications using LangChain and open-source LLMs. Skilled in deploying containerized, production-style APIs with FastAPI and Docker. Strong foundation in IoT and hardware-software integration.

TECHNICAL SKILLS

AI & ML: YOLOv8, EasyOCR, LangChain, ChromaDB, RAG, Sentence-Transformer Embeddings, Deep Learning
LLM & APIs: Groq (LLaMA 3), OpenAI API, Anthropic API, Prompt Engineering, FastAPI, Flask
Languages: Python, JavaScript (ES6+), SQL, Kotlin, Java, C++
Deployment: Docker, Git/GitHub, Linux, Uvicorn
IoT & Hardware: ESP32, Arduino Uno, Sensor Integration, Firmware Development

KEY PROJECTS

Traffic Violation RAG Assistant *LangChain · ChromaDB · Groq LLaMA 3 · FastAPI · Docker*

- Built a RAG system enabling natural-language queries over a 250-record traffic violation database using sentence-transformer embeddings (all-MiniLM-L6-v2) and ChromaDB vector store — achieved 90%+ top-4 retrieval relevance with under 2s response time via Groq inference.
- Deployed a FastAPI service with /query and /ingest endpoints, fully containerized with Docker for one-command reproducible deployment.

Traffic Violation Detection System *YOLOv8 · EasyOCR · Flask*

- Engineered a real-time pipeline using YOLOv8 and EasyOCR to detect vehicles and recognize license plates from live camera feeds.
- Built a centralized Flask dashboard to visualize live detections and manage historical violation logs.

Smart Soil Health & Compost Monitor *ESP32 · IoT · Cloud*

- Designed an ESP32-based IoT system monitoring soil pH, moisture, and temperature with cloud-based alerts for real-time composting optimization.

Expense Tracker Mobile App *Kotlin · SQLite · Android*

- Built a native Android app in Kotlin for daily financial tracking with interactive charts and SQLite-backed local data persistence.

CERTIFICATIONS

- Mastering Retrieval-Augmented Generation (RAG) — Udemy (April 2025)
- Deep Learning — Simplilearn (May 2025)
- Learn Python Masterclass — Udemy (March 2025)

EDUCATION

BCA — Artificial Intelligence & Machine Learning Jun 2023 – May 2026 (Expected) Presidency University, Bangalore

PCME — Pre-University Aug 2021 – Mar 2023 St. Joseph's PU College, Bangalore